higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ELECTROTECHNICS N6

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This marking guideline consists of 10 pages.

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QUESTION 1: DC MACHINES

1.1 It is working at maximum efficiency when the constant losses✓ and the armature copper losses are equal.✓ (2)

1.2
- Armature copper loss
- Interpole copper loss
- Series winding copper loss
- Brush resistance loss
- Compensating winding copper loss (Any 3 × 1) (3)

1.3 The speed of a DC motor increases with an increase of armature voltage✓ but not in direct proportion.✓ (2)

1.4 1.4.1 The output power

$$P_{motor} = (W - P) \times 9,81 \times \pi Dn$$
$$¿ (35 - 10) \times 9,81 \times \pi \times \frac{40}{100} \times \frac{950}{60}$$
$$¿ 4\ 879,679\ W$$ (2)

1.4.2 The efficiency

$$P_{motor\ input} = V_{motor} \cdot I_{motor} = 200 \times 30 = 6\,000\ W$$

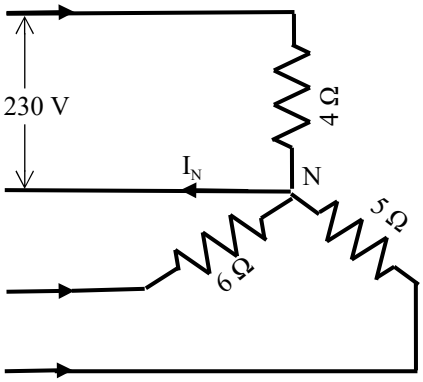
$$\%\ Efficiency = \frac{P_{motor\ output}}{P_{motor\ input}} \times 100\%$$
$$¿ \frac{4879,679}{6\,000} \times 100\%$$

$$¿ 81,328\%$$ (4)

[13]

QUESTION 2: AC CIRCUIT THEORY

2.1 GIVEN: A 3-phase, star-connected system



2.1.1 The current flowing in each phase

$$I_{R4\Omega}=\frac{V}{R_4}=\frac{230}{4}=57,\ 5\ \ A$$

$$I_{R5\Omega}=\frac{V}{R_5}=\frac{230}{5}=46\ \ A$$

$$I_{R6\Omega}=\frac{V}{R_6}=\frac{230}{6}=38,\ 333\ \ A$$ (6)

2.1.2 The neutral current

$$I_{X\text{-}comp}=I_{R5\Omega}\cos 30°-I_{R6\Omega}\cos 30°$$

$$¿46\cos 30°\text{-}38,\ 333\cos 30°$$

$$=6,\ 64\ \ A$$

$$I_{Y\text{-}comp}=I_{R4\Omega}-I_{R5\Omega}\sin 30°-I_{R6\Omega}\sin 30°$$

$$=57,\ 5\text{-}46\sin 30°\text{-}38,\ 333\sin 30°$$

$$¿15,\ 334\ \ A$$

$$I_N=\sqrt{I_{X\text{-}comp}{}^2+I_{Y\text{-}comp}{}^2}$$

$$¿\sqrt{6,\ 64^2+15,\ 334^2}$$

$$¿16,\ 71\ \ A$$ (6)

2.1.3 The total power absorbed

$$P=P_{R4\Omega}+P_{R5\Omega}+P_{R6\Omega}=230(57,\ 5+46+38,\ 333)\qquad =32\,621{,}59\ \ W$$ (2)

2.2 A 3-phase balanced load is an electrical circuit in which the loads✓ connected across three phases, are identical.✓ (2)

[16]

QUESTION 3: TRANSFORMERS

3.1
- Star-delta
- Delta-star
- Star-star
- Delta-delta (4)

3.2 Given: A single-phase transformer

S = 650 kVA η_{max} = 95%
pf = 0,8 lagging k = 85%

3.2.1 The iron loss at maximum efficiency

Method 1

$$\eta_{max} = \frac{k \cdot S \cdot \cos\theta}{k \cdot S \cdot \cos\theta + 2P_o} \times 100\%$$

$$0,95 = \frac{0,85 \times 650 \times 0,8}{(0,85 \times 650 \times 0,8) + 2P_o} = \frac{442}{442 + 2P_o}$$

$$2P_o = \frac{442}{0,95} - 442 = 23,263$$

$$P_o = \frac{23,263}{2} = 11,632\ kW$$

OR

Method 2

$$k = \sqrt{\frac{P_o}{P_{sc}}}$$

$$\therefore\ P_o = k^2 \cdot P_{sc} = 0,85^2 \times P_{sc} = 0,723P_{sc}$$

$$\eta_{max} = \frac{k \cdot S \cdot \cos\theta}{k \cdot S \cdot \cos\theta + P_o + k^2 \cdot P_{sc}} \times 100\%$$

$$0,95 = \frac{0,85 \times 650 \times 0,8}{(0,85 \times 650 \times 0,8) + 0,723P_{sc} + 0,85^2 P_{sc}}$$

$$0,95 = \frac{442}{442 + 1,446P_{sc}}$$

$$1,446P_{sc} = \frac{442}{0,95} - 442 = 23,263$$

$$P_{sc} = \frac{23,263}{1,446} = 16,088\ kW$$

$$P_o = k^2 \cdot P_{sc} = 0,85^2 \times 16,088 = 11,624\ kW$$ (4)

3.2.2 The copper loss at maximum efficiency

$$k = \sqrt{\frac{P_o}{P_{sc}}}$$

$$P_{sc} = \frac{P_o}{k^2} = \frac{11,\ 624}{0,\ 85^2} = 16,\ 089\ \ kW$$ (2)

3.2.3 The copper loss at full load

$$P_{sc} = \frac{P_{sc}}{k} = \frac{16,\ 089}{0,\ 85} = 18,\ 928\ \ kW$$ (1)

3.2.4 The full-load efficiency

$$\eta_{max} = \frac{S \cdot \cos\theta}{S \cdot \cos\theta + P_o + P_{sc}} \times 100\,\%$$

$$= \frac{650 \times 0,\ 8}{(650 \times 0,\ 8) + 11,\ 624 + 16,\ 089} \times 100$$

$$= 94{,}94\,\%$$ (2)

[13]

QUESTION 4: AC MACHINES - GENERATORS

4.1 Armature reaction is the influence of the stator ampere-turns ✓ on the value and distribution of the magnetic flux✓ in the air gaps between the poles and stator core.✓ (3)

4.2 Given: A three-phase, star-connected alternator

V_L = 3 600 V slots/pole/phase = 3

f = 50 Hz Z/slot = 20

N = 500 r/min K_d = 0,96

4.2.1 The number of poles

$$f = \frac{N \cdot p}{60}$$

$$p = \frac{60\ f}{N} = \frac{60 \times 50}{500} = 6\ \ pairs$$

$$\therefore number\ \ of\ \ poles = 6 \times 2 = 12$$ (2)

4.2.2 The useful flux per pole

$$E_L = \frac{V_L}{\sqrt{3}} = \frac{3\ 600}{\sqrt{3}} = 2\,078,\ 461\ V$$

n = Number of slots per pole/phase
Number of slots per pole = n × phase✓ = 3 × 3 = 9✓
Number of slots per pole = Number of slots/Number of poles
Number of slots = Number of slots per pole × Number of poles ✓
= 9 × 12
= 108✓

Z = Total conductors in series per phase

= (108 × 20) / 3 ✓

= 720✓

$$E = 2\,k_f\,k_d\,k_p\,f\Phi Z$$

$$\Phi = \frac{E}{2\,k_f\,k_d\,k_p\,fZ}$$

$$\Phi = \frac{2\,078{,}461}{2 \times 1{,}11 \times 0{,}96 \times 1 \times 50 \times 720}$$

$$\Phi = 0{,}027\ 090\ Wb$$ ✓ (9)

4.3 Distribution factor – is the ratio of the actual EMF✓ obtained to the EMF that would be concentrated in one slot.✓ (2)
[16]

QUESTION 5: AC MACHINES – SYNCHRONOUS MOTORS

5.1 The V-curves of a synchronous motor shows how the armature current✓ varies with the field current✓ when the input is kept constant. ✓ (3)

5.2 Given: A three-phase, star-connected synchronous motor
S = 7 000 VA P = 5,5 kW
V = 450 V Z_s = (0,35 + j3,5) ohms/phase

5.2.1 The power factor

$$\cos\theta = \frac{P}{S} = \frac{5\ 500}{7\ 000} = 0{,}786\ \ leading$$ (1)

5.2.2 The phase current

Method 1

$P = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos\theta$

$\therefore\ I_L = \dfrac{P}{\sqrt{3} \cdot V_L \cdot \cos\theta}$

¿ $\dfrac{5\,500}{\sqrt{3} \times 450 \times 0,\ 786}$

$= 8,\ 978\ A$

Method 2

$S = \sqrt{3} \cdot V_L \cdot I_L$

$\therefore\ I_L = \dfrac{S}{\sqrt{3} \cdot V_L} = \dfrac{7\,000}{\sqrt{3} \times 450}$

$= 8,\ 981\ A$

(2)

5.2.3 The EMF to which the machine is excited

$\cos\theta = 0,\ 786$

$\therefore\ \theta = \cos^{-1} 0,\ 786 = 38,\ 187°$

$V_{ph} = \dfrac{V_L}{\sqrt{3}} = \dfrac{450}{\sqrt{3}} = 259,\ 808\ V$

Method 1

$Z_S = R_a + J\,X_S = 0,\ 35 + j\,3{,}5$

$= 3,\ 517\ 84,\ 289°$

$E_r = I\,Z_S = 8,\ 981\ 38,\ 187° \times 3,\ 51$

Method 2

$I\,R_a = 8,\ 981 \times 0,\ 35$

¿ $3,\ 143\ V$

$I\,X_S = 8,\ 981 \times 3,\ 5$

¿ $31,\ 434\ V$ ✓

$E_{ph} = E_r - V_{ph}$

¿ $31,\ 586\,122,\ 476° - 259,\ 808\,0$

$= 278,\ 048\,174,\ 501°\ V$

OR

$E_{ph} = V_{PH} + I\,R_a + I\,X_S$

$= 259,\ 808 - 38{,}187° + 3,\ 143\,180° + 31,\ 434 - 90°$

¿ $278,\ 051 - 43,\ 687°\ V$

$E_L = \sqrt{3} \cdot E_{ph} = \sqrt{3} \times 278,\ 048 = 481,$ $E_L = \sqrt{3} \cdot E_{ph} = \sqrt{3} \times 278,\ 051$

$= 481,\ 598\ V$

(6)

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5.2.4 The load angle in electrical degrees

Method 1	Method 2
α = 180º – 161,373º	α = 56,815º – 38,187º
= 18,627º ✓	= 18,628º ✓

(1)
[13]

QUESTION 6: AC MACHINES – INDUCTION MOTORS

6.1 Given: A 3-phase, 50-Hz induction motor
p = 8-pole T_{MAX} = 185 Nm
R_r = 0,3 Ω S_{MAX} = 5%
N = 360 r/min

The torque developed by the rotor

$$N_s = \frac{60\,f}{p} = \frac{60 \times 50}{8} = 375\ \ r/min = 6{,}25\ \ r/s$$

$$S_{fractional} = \frac{N_s - N}{N_s} = \frac{375 - 360}{375} = 0{,}\ 04$$

$$R_2 = S\,X_0 \quad \therefore\ X_0 = \frac{R_2}{S} = \frac{0{,}3}{0{,}04} = 7{,}5\ \ \Omega$$

$$T_{MAXIMUM} = \frac{3 \cdot S \cdot E_0^{\,2}}{2 \cdot \pi \cdot N\left(R_2^{\,2} + \left[S \cdot X_0\right]^2\right)}$$

$$= \frac{3 \times 0{,}\ 04 \times E_0^{\,2}}{2 \times \pi \times \frac{360}{60} \times \left(0{,}\ 3^2 + \left[0{,}\ 04 \times 7{,}\ 5\right]^2\right)}$$

$$= 0{,}017\ \ 684 \times \ \ E_0^{\,2}$$

$$\therefore\ E_0 = \sqrt{\frac{T_{max}}{0{,}017\ \ 684}} = \sqrt{\frac{185}{0{,}017\ \ 684}} = 102{,}281\ \ V$$

$$T_{START} = \frac{3 \cdot S \cdot E_0^{\,2}}{2 \cdot \pi \cdot n\left(R_2^{\,2} + \left[S \cdot X_0\right]^2\right)}$$

$$¿\ \frac{3 \times 0{,}\ 05 \times (102{,}281)^2}{2 \times \pi \times 6{,}25 \times \left(0{,}\ 3^2 + \left[0{,}\ 05 \times 7{,}\ 5\right]^2\right)} = 173{,}267\ \ Nm$$

(9)

6.2 Slip is the difference between the speed of the rotating field✓ and the actual speed or the rotor.✓ (2)

6.3
- Actual measurement of motor speed
- Comparing rotor and stator supply frequencies
- Stroboscopic method (3)

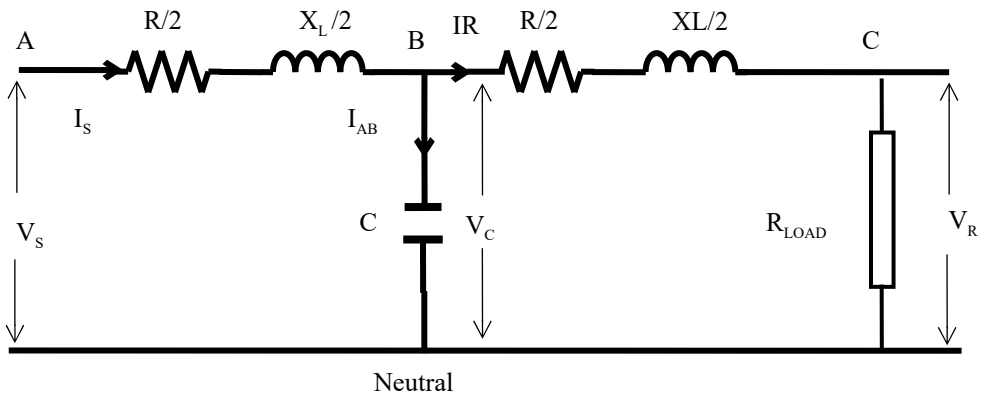
6.4 6.4.1 Decreases
6.4.2 Increases

(2 × 1) (2)
[16]

QUESTION 7: GENERATION AND DISTRIBUTION OF AC

Given: A 3-phase transmission line

S = 20 MW	C = 0,08 mF/km	L = 100 km
V_L = 110 kV	R = 0,2 Ω/phase/km	X_L = 0,4 Ω/phase/km
f = 50 Hz	pf = 0,9 lagging	



7.1 The sending end current

$$R/2 = (0,\ 2\times 100)/2 = 10\ \Omega$$
$$\frac{X_L}{2} = \frac{0,\ 4\times 100}{2} = 20\ \Omega$$
$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 0,\ 08\times 10^{-3}} \times 100 = 3978{,}874\ \Omega$$
$$V_R = V_L/\sqrt{3} = 110\ 000/\sqrt{3} = 63\ 508{,}530°\ V$$

Method 1

$$P = \sqrt{3}\cdot V_L \cdot I_L \cdot \cos\theta$$
$$I_L = P/\sqrt{3}\cdot V_L \cdot \cos\theta$$
$$¿ 20\times 10^6/\sqrt{3}\times 110\ 000\times 0,\ 9$$
$$= 116,\ 636\text{-}25,\ 842°\ A$$

Method 2

$$S = \frac{P}{\cos\theta} = \frac{20\times 10^6}{0,\ 9} = 22,\ 222\ MW$$
$$S = \sqrt{3}\cdot V_L \cdot I_L$$
$$\therefore I_L = S/\sqrt{3}\cdot V_L$$
$$¿ 22,\ 222\times 10^6/\sqrt{3}\times 110\times 10^3$$
$$= 116,\ 636\text{-}25,\ 842°\ A \qquad (4)$$

7.2 The sending end voltage

$$Z_{BC} = Z_{AB} = \frac{R}{2} + \frac{jX_L}{2} = 10 + j20 = 22,\ 36163{,}435°\ \Omega$$

$$V_{BC} = I_R . Z_{BC} = 116,\ 636\text{-}25,\ 842° \times 22,\ 36163,\ 432°$$
$$= 2\ 608,\ 09837,\ 59°\ V$$

$$V_C = V_{BC} + V_R = 2608,\ 09837,\ 59° + 63508,\ 530°$$
$$= 65\ 594,\ 4731,\ 39°\ V$$

$$I_C = V_C/Z_C = 65594,\ 4731,\ 39°/3978,\ 874\text{-}90° = 16,\ 48691{,}39°\ A$$

$$I_S = I_C + I_R = 16,\ 48691{,}39° + 116,\ 636\text{-}25{,}842°$$
$$= 110,\ 073\text{-}18,\ 189°\ A$$

$$V_{AB} = I_S . Z_{AB} = 110,\ 073\text{-}18,\ 189° \times 22,\ 36163,\ 435°$$
$$= 2461,\ 34245,\ 246°\ V$$

$$V_S = V_{AB} + V_C = 2461,\ 34245,\ 246° + 65594{,}4731,\ 39°$$
$$= 67\ 390,\ 8862,\ 84°\ V$$

$$V_L = \sqrt{3}.V_S = \sqrt{3}\times 67390,\ 886$$
$$= 116\ 724,\ 439\ V \qquad (8)$$

7.3 The power factor

$$Power\ \ factor = \cos(29{,}556° + 9,\ 291°) = \cos 38,\ 847° = 0,\ 779\ \ lagging \qquad (1)$$

[13]

TOTAL: 100

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